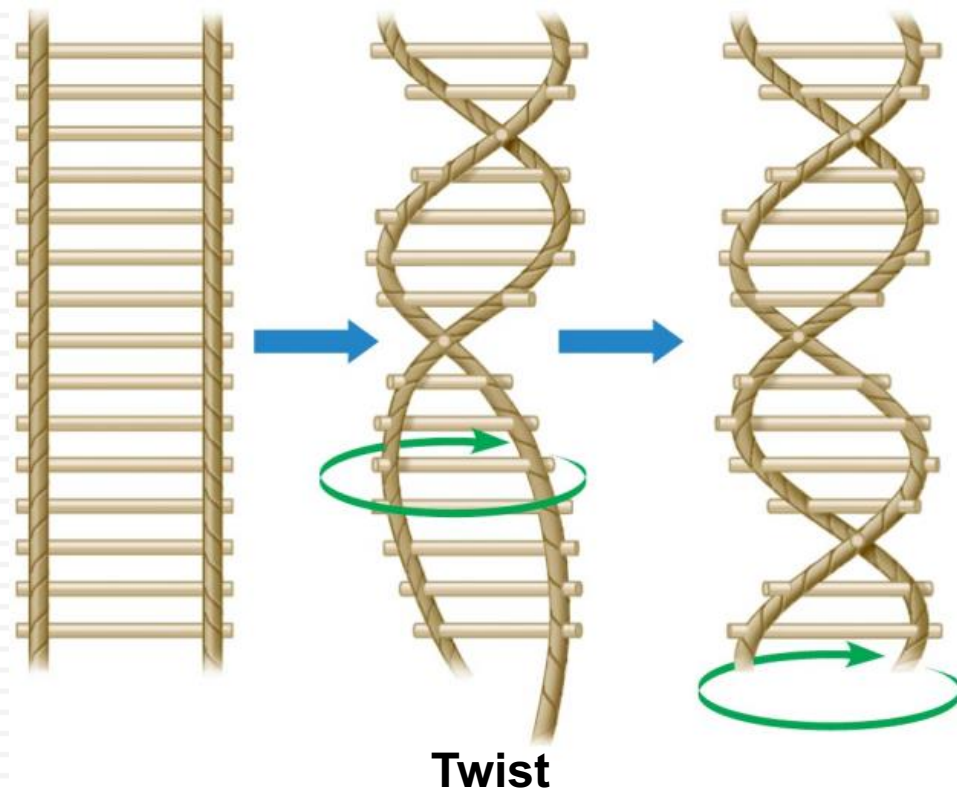


UNIT 4: MOLECULAR GENETICS

4.2 – DNA STRUCTURE

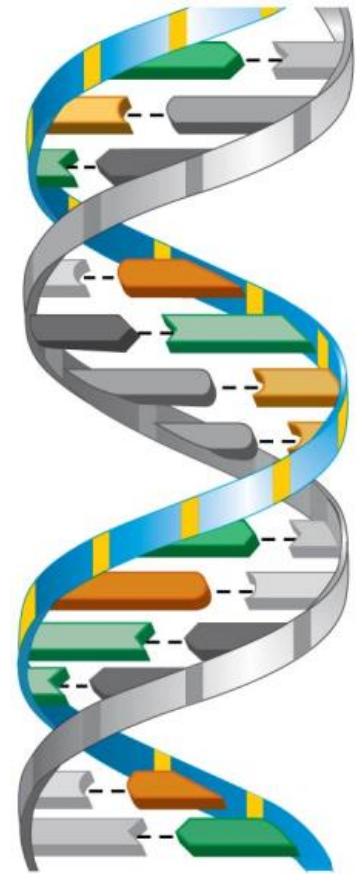
Biology 9 Mr. Queenan
Ch. 13.2 (p. 418-423)

DNA Structure



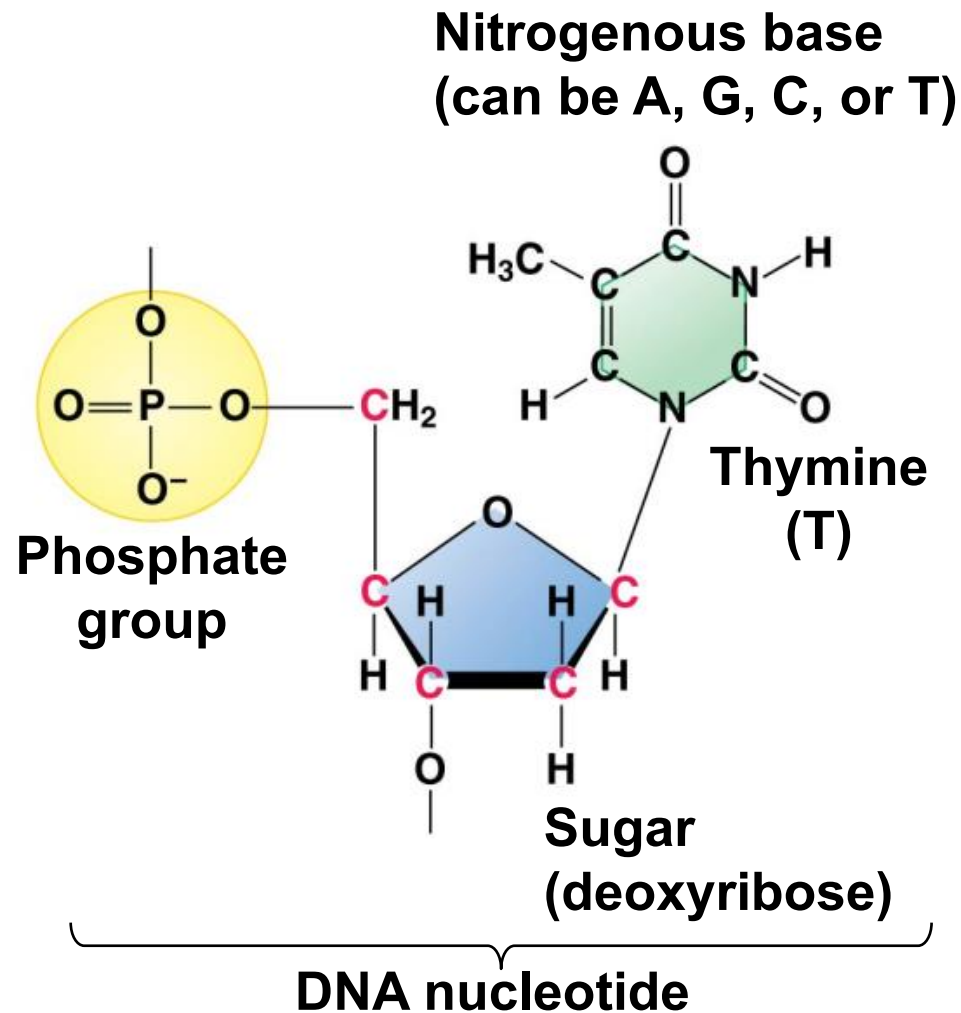
DNA Nucleotides

- **DNA:** nucleic acid composed of 2 long chains (strands) of repeating subunits: nucleotides
 - ▣ One molecule of DNA is composed of 2 strands
- **Nucleotide:** composed of 3 parts
 - ▣ 5- carbon sugar (deoxyribose)
 - ▣ Phosphate group (PO_4)
 - ▣ Nitrogenous base (contain C, N and acts as a base – accepts H ions)
 - Adenine, thymine, guanine, cytosine
- Sugar and phosphate molecules alternate to form “handrails of staircase”; also known as sugar phosphate backbone
- Nucleotides of each strand connected by covalent bonds between phosphate group of one nucleotide and sugar of next
 - ▣ Phosphodiester bonds
- Each ‘turn’ of the helix contains ~10 nucleotides



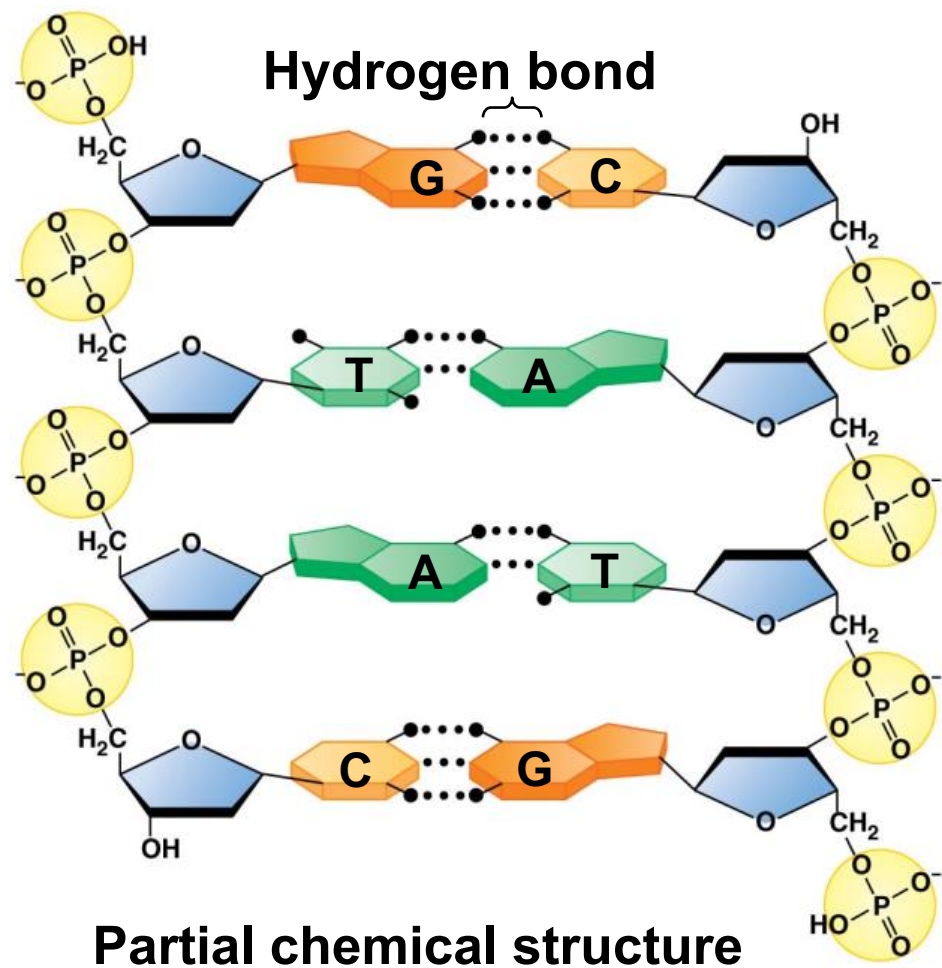
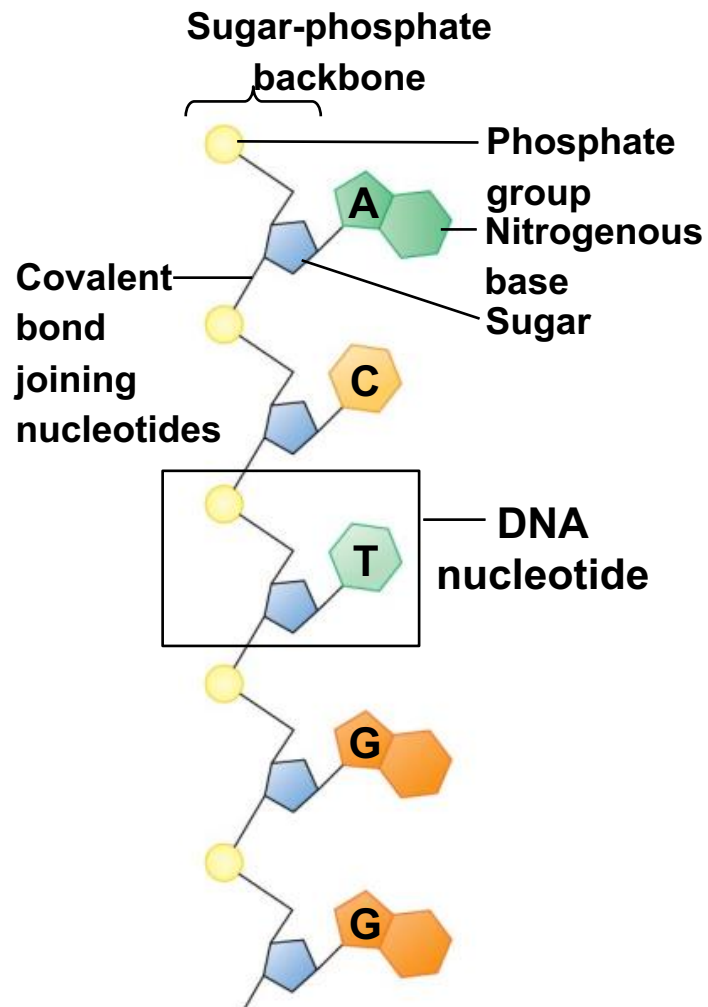
**A DNA
double helix**

DNA Nucleotides



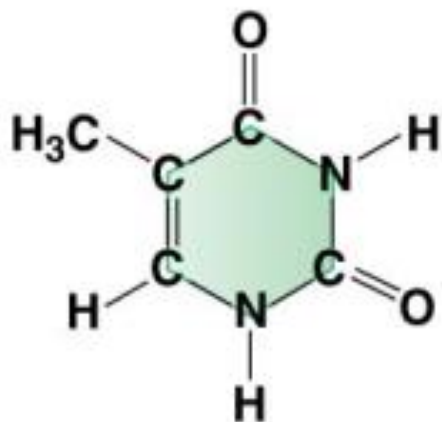
DNA Bonding

- Sugar-phosphate backbone on outside of molecule; bases face toward center of molecule
 - ▣ 'Rungs' or 'steps' of the staircase
- Bases face each other; form hydrogen bonds with bases on other strand
- Adenine (A) always pairs with Thymine (T)
 - ▣ Forms 2 hydrogen bonds
- Guanine (G) always pairs with Cytosine (C)
 - ▣ Forms 3 hydrogen bonds
- Hydrogen bonds between bases hold the 2 strands of the DNA molecule together

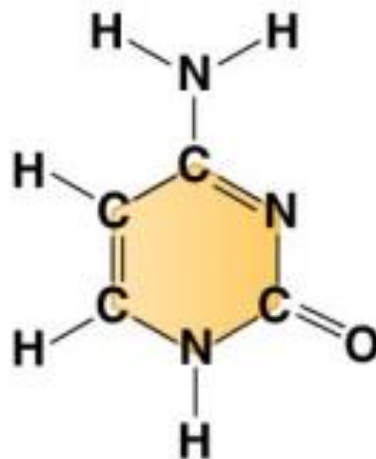


Nitrogenous Bases

- Sugar/phosphate component is same among all nucleotides
- Nitrogenous bases differ:
 - ▣ Adenine (A)
 - ▣ Guanine (G)
 - ▣ Cytosine (C)
 - ▣ Thymine (T)
- Each base has a different chemical structure; all composed of C rings with N groups
- Bases with double-ring structure → **purines**
 - ▣ **A**denine; **G**uanine
- Bases with single ring structure → **pyrimidines**
 - ▣ **T**hymine; **C**ytosine

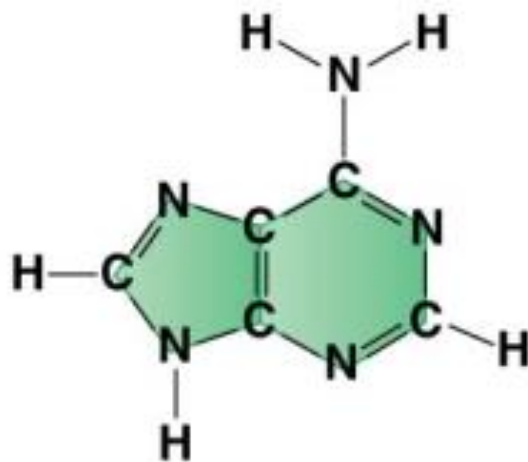


Thymine (T)

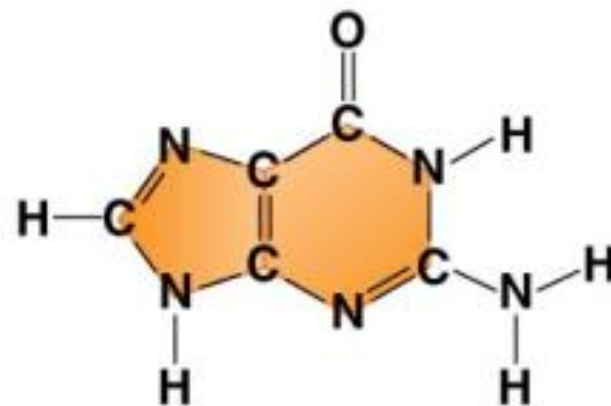


Cytosine (C)

Pyrimidines



Adenine (A)

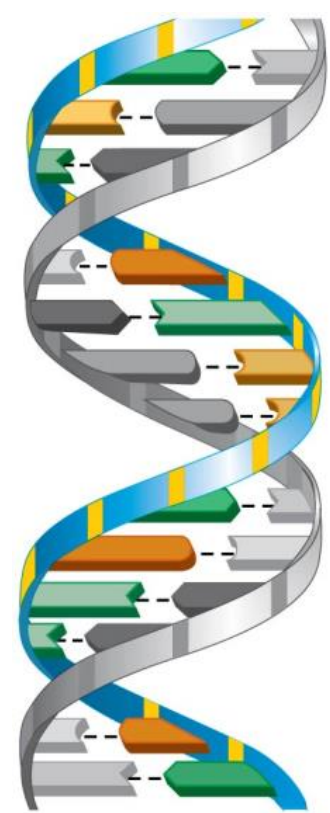


Guanine (G)

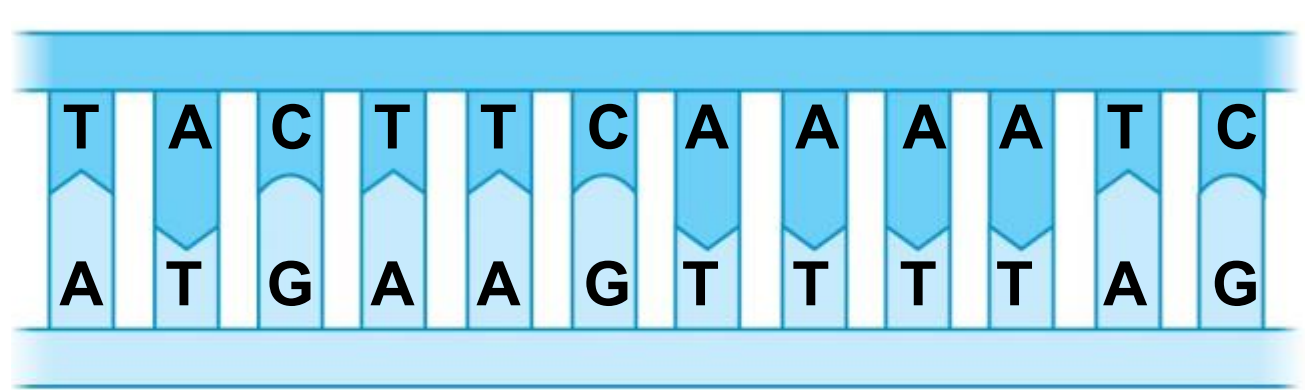
Purines

Complementary Bases

- American biochemist Chargoff observed in a sample of DNA:
 - ▣ % of A equals % of T; % of C equals % of G
- Led to development of base pairing rules, aka Chargoff's rules
 - ▣ C on one strand always pairs with G on other
 - ▣ A on one strand always pairs with T on other
 - ▣ Pairs called complementary base pairs
- Pairs contain one purine; one pyrimidine
- Because of this:
 - ▣ Order of bases on one strand is complementary to order of bases on other strand
 - ▣ Order of bases known as base sequence, or DNA sequence
- Complementary base pairing important to DNA structure
 - ▣ Hydrogen bonding holds strands together
 - ▣ Complementary structure explains how DNA can be replicated before cell division
 - One strand of DNA serves as template for formation of new complementary strand



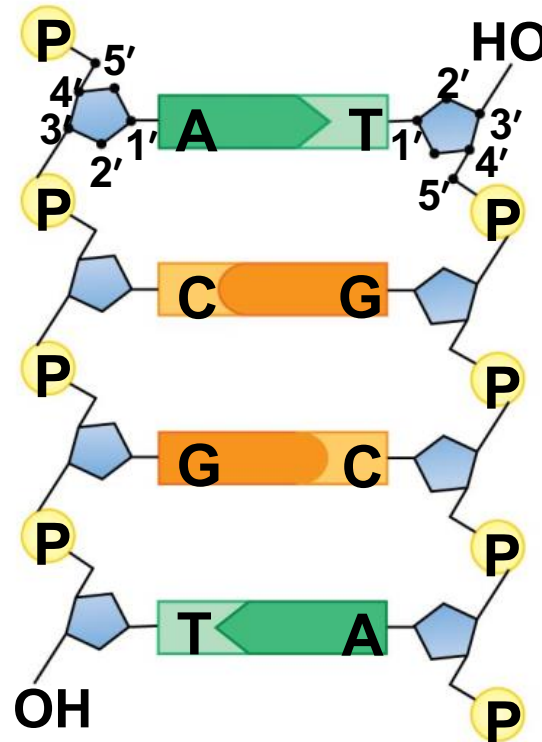
Ribbon Model



Straight Ladder Model



Space-filling Model



Stick Model

CGATTAGCTAGCCCGAT
GCTAATCGATCGGGCTA
Sequence Model

Eukaryotic Chromosomes

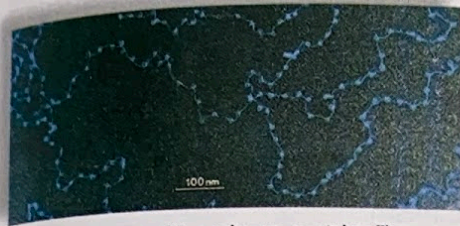


Eukaryotic Chromosome

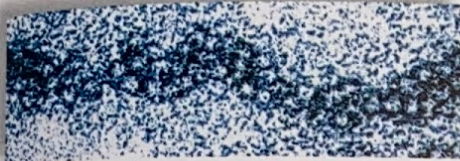
- Made of a single double helix DNA molecule
 - ▣ More than 50% of composition is protein
 - Some concerned with DNA/RNA synthesis
 - Large majority are **histones**
 - Play a structural role in the chromosome
- 5 primary types of histone proteins
 - ▣ H1, H2A, H2B, H3, & H4
 - H3 & H4 structure highly conserved across organisms
 - Essential function for survival conserved across organisms

Histones

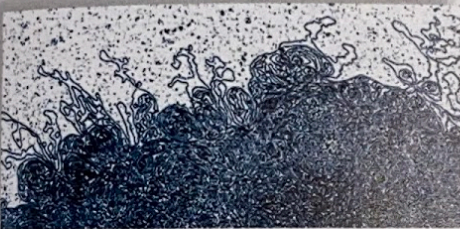
- Human cells contain upwards of 2m of DNA
 - ▣ Must be packed down to fit into 5µm diameter nucleus
 - ▣ Histones responsible for packaging DNA into nucleus
- DNA wound at intervals around cores of 8 histones
 - ▣ 2 copies of each H2A, H2B, H3, & H4
 - ▣ Called a **histone octet**
 - ▣ Appear like beads on a string
 - Each bead known as a **nucleosomes** – joined by “linker DNA”
- String compacts in zig-zag folded structure between nucleosomes
 - ▣ H1 histone plays role in assisting this folding process



a. Nucleosomes ("beads on a string")



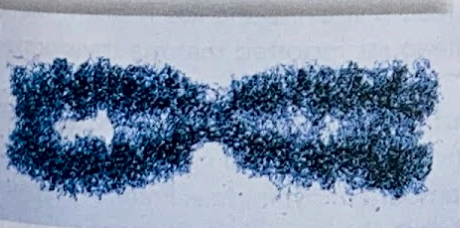
b. 30-nm fiber



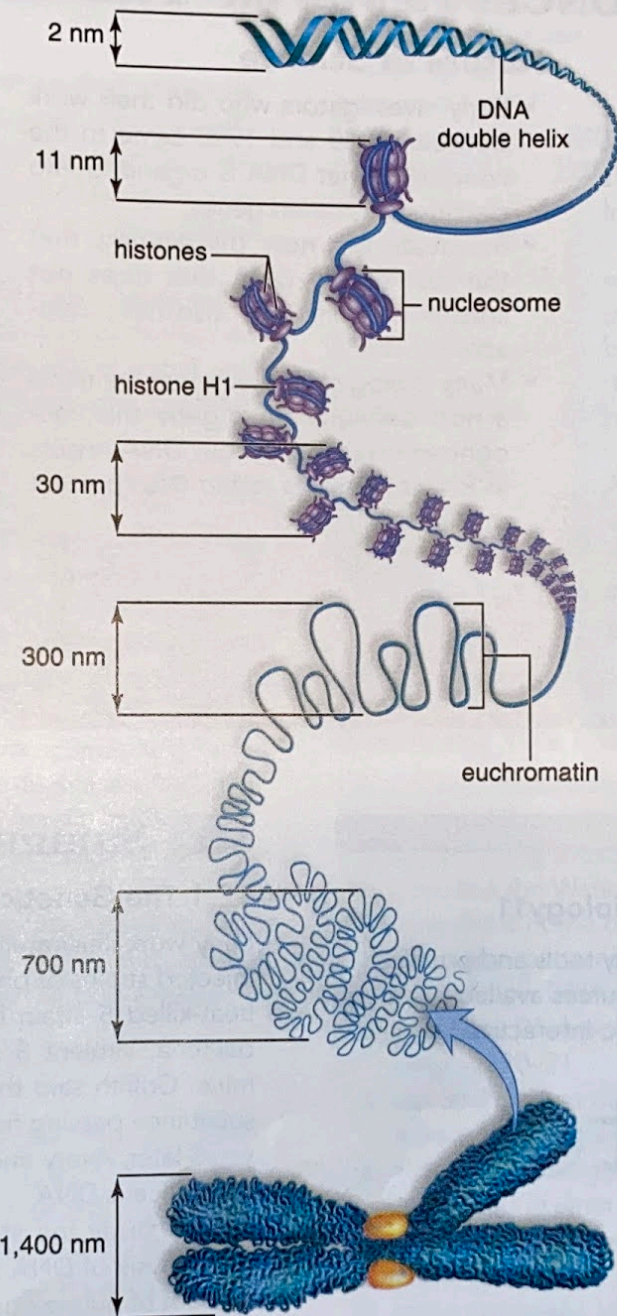
c. Radial loop domains



d. Heterochromatin



e. Metaphase chromosome



1. Wrapping of DNA around histone proteins.

2. Formation of a three-dimensional zigzag structure via histone H1 and other DNA-binding proteins.

3. Loose coiling into radial loops.

4. Tight compaction of radial loops to form heterochromatin.

5. Metaphase chromosome forms with the help of a protein scaffold.

Chromatin

- Zig-zags continue folding back and forth into radial loops
- These loose coiled loops are known as **euchromatin**
 - ▣ Region of DNA with active chromatin containing genes that are being transcribed
- DNA that gets further compacted forms **heterochromatin**
 - ▣ Region of DNA considered inactive because genes contained in it are infrequently transcribed
 - ▣ Usually dark-stained fibers in the nucleus when observed using microscopy
- Most chromosomes exhibit both levels of compaction, depending on which regions of the chromosome is being used more frequently
- Regulating the level of compaction is an important method of controlling gene expression in a cell